

RAMCO INSTITUTE OF TECHNOLOGY
Department of Mechanical Engineering
Academic Year 2021 – 22

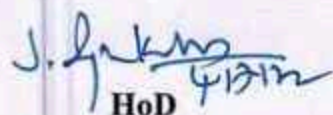
Ref.: RIT/Mech./PA&QIC/05

Date: 04.07.2022

Circular

The 5th Program Assessment & Quality Improvement Committee Meeting to review the progress of Academic Year 2021-22 (Even Semester) is planned on **08.07.2022 (Friday)** from **2.30 PM to 5:00 PM** in Mechanical Seminar Hall. All PA&QIC Members and other faculty members are invited to attend the meeting. The agenda for the meeting is listed below.

Sl. No.	Topics to present	Faculty In charge	Time
	Welcome Address – Mr.R.Arun Kumar, AP/Mech.		2:30PM - 2:32PM
1.	Review of Previous meeting (4 th PA&QIC)	Dr.P.Suresh Kumar, ASP/Mech.	3:02PM - 3:07PM
2.	(i) New initiatives taken for quality improvement in Teaching and Learning (ii) Feedback analysis [Student / Employee / Recruiters /Parents / Alumni]	Mr.G.Prabu ram, AP(SG)/Mech.	3:07PM - 3:20PM
3.	Impact analysis of Internship towards attainment	Mr.S.Kathiravan, AP/Mech.	3:20PM - 3:30PM
4.	Comparative analysis of CO attainment for Design / Manufacturing / Thermal domains	Dr.J.Jabinth, AP/Mech.	3:30PM - 3:40PM
5.	Proposed methodologies to set target for CO attainment i) Design Domain and ii) Manufacturing Domain	(i)Mr.J.Jerold John Britto, AP (SG)/Mech. (ii)Mr.S.Maharajan, AP/Mech.	3:40PM - 3:55PM
6.	Student's performance in Co-Curricular and Extra Curricular Activities	Mr.S.Valai Ganesh, AP/Mech.	3:55PM - 4:00PM
7.	Faculty Progress	Mr.N.L.Sujin, AP/Mech.	4:00PM - 4:15PM
8.	Budget utilization (AY2021-2022)	Mr.M.Sivagaminathan alias Balaji, AP(SG)/Mech.	4:20 PM - 4:30PM
9.	Continuous Improvement – department progress	Mr.L.Karthikeyan, AP/Mech.	4:30PM - 4:40PM
	Feedback and Suggestion from members of the committee		4:40PM - 4:55PM
	Vote of Thanks – Dr.V.Sivakumar, ASP/Mech.		4:55PM - 5:00PM


HoD


Vice Principal


Principal

Copy to,

- 1) All PA&QIC Members and faculty members through email
- 2) File



RAMCO INSTITUTE OF TECHNOLOGY

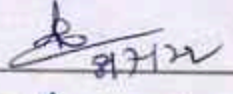
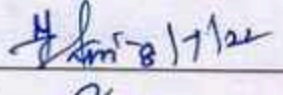
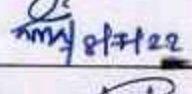
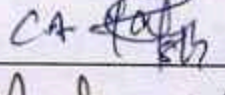
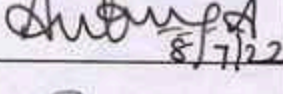
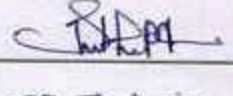
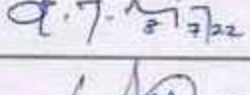
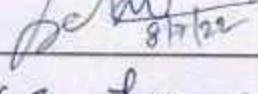
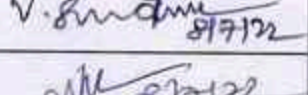
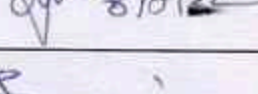
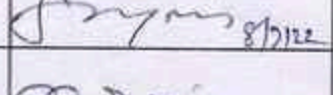
Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Date: 08.07.2022

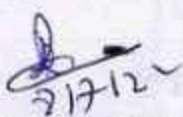
5th Program Assessment & Quality Improvement Committee Meeting

Department of Mechanical Engineering

Attendance Sheet

S.No.	Name of the faculty member	Designation	Signature
1	S. Rajakarunakaran	Prof	
2	M. ASHOK KUMAR	Asst Professor (SQ)	
3	R. RAMALAKSHMI	AP / ECE	
4	C. A. YOGARAJA	AP / CSE	
5	Arun kumar A	AP / EEE	
6	O. SENTHILKUMAR	ASP / Chemistry	
7	S. MAHARAJAN	AP / MECH	
8	G. Selvaraj	AP / Maths	
9	V. Sivakumar	ASP / Mech.	
10	S. Godwin Bomabas	Asp / Mech	
11	P. RABHAKARAN	AP / Mech	
12	C. Gururaj	AP(SQ) / Mech	

S.No.	Name of the faculty member	Designation	Signature
13.	J. Jeyaraj Jeyaraj Berito	AP/SG	J. Jeyaraj
14.	R. VENKATESH	AP/Mech	R. Venkatesh
15	M. RAMAR	AP/Mech	M. Ramar
16	S. Lathavaleen	AP/Mech	S. Lathavaleen
17	S. VALDI GANESH	AP/mech	S. Valdi Ganesh
18	Santhana Maruthu Pandian M	AP/mech	Santhana Maruthu Pandian M
19.	R. Arun Kumar	AP/Mech	R. Arun Kumar
20	T. Jabinth	AP/mech	T. Jabinth
21	L. KARUNESWARAN.	AP/Mech	L. Karuneswaran.
22	P. Suresh Kumar	AP/mech	P. Suresh Kumar


 21/12/20
 HoD - Mechanical



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering
Academic Year 2021 – 2022 (Even Semester)

Minutes of Meeting – 5th Program Assessment & Quality Improvement Committee (PA&QIC)

Brief Introduction about PA&QIC

Objective: To enhance the quality of the program and to ensure continuous improvement in terms of Curriculum, Teaching Learning process (TLP), Student Performance, Faculty Contributions, Infrastructural facilities, Student support system and Industry interactions by effectively involving the stakeholders

Meeting frequency: Twice per year

Roles and Responsibilities:

Convener: Formulating the agenda and convene the meeting

Member & Representative: Share their views and suggestion for continuous improvement.
Monitor the progress of the department.

Deliverables: Submit report to Department Advisory Committee on effectiveness of program implementation in terms of each process analysis, action taken and impact.

Date of Meeting : 08.07.2022 (Saturday)
Time of Meeting : 2:30 PM to 5:00 PM
Reference : RIT/Mech./PA&QIC/05
Venue : Mechanical Seminar Hall
Members Present :

Sl. No.	Name of the Faculty Member	Designation	Responsibility
1.	Dr.S.Rajakarunakaran	Professor and Head / Mechanical	Convener
2.	Mr.M.Ashok Kumar	AP(SG) / Mechanical	Member
3.	Mr.S.Maharajan	Assistant Professor / Mechanical	Member
4.	Mr.R.Prabhakaran	Assistant Professor / Mechanical	Member
5.	Mr.C.A.Yogaraja	Assistant Professor / CSE	Member
6.	Mr.A.Arun Kumar	Assistant Professor (SG) / EEE	Member
7.	Mrs.R.Ramalakshmi	Assistant Professor (SG) / ECE	Member
8.	Dr.G.Selvaraj	Assistant Professor / Maths	Member
9.	Dr.P.Suresh Kumar	Associate Professor / Mechanical	Representative
10.	Dr.V.Sivakumar	Associate Professor / Mechanical	Representative

Sl. No.	Name of the Faculty Member	Designation	Responsibility
11.	Dr.S.Godwin Barnabas	Associate Professor / Mechanical	Representative
12.	Mr.J.Jerold John Britto	AP(SG) / Mechanical	Representative
13.	Mr.M.Sivagaminathan alias Balaji	Assistant Professor (SG)	Representative
14.	Mr.S.Valai Ganesh	AP / Mechanical	Representative
15.	Mr.C.Gururaj	AP(SG) / Mechanical	Representative
16.	Dr.J.Jabinth	AP / Mechanical	Representative
17.	Mr.L.Karthikeyan	AP / Mechanical	Representative
18.	Mr.G.Prabu ram	AP(SG) / Mechanical	Representative
19.	Mr.M.Santhana Maruthu Pandian	AP / Mechanical	Representative
20.	Mr.R.Arun Kumar	AP / Mechanical	Representative
21.	Mr.R.Venkatesh	AP / Mechanical	Representative
22.	Mr.M.Ramar	AP / Mechanical	Representative
23.	Mr.S. Kathiravan	AP / Mechanical	Representative

Leave / on duty:

Sl. No.	Name of the Faculty Member	Designation	PAQIC	CL/OD
1.	Mrs.R.Kalaimani	AP/Civil	Member	CL
2.	Dr.M.Lakshmanan	AP(SG)/Mech.	Representative	OD
3.	Mr.T.Selvasundar	AP/Mech.	Representative	CL

The agenda of the meeting

1. Review of Previous meeting (4th PA&QIC)
2. Teaching and Learning:
 - (i) New initiatives taken for quality improvement in Teaching and Learning
 - (ii) Feedback analysis [Student / Employee / Recruiters /Parents / Alumni] Impact analysis of internship and industry training
3. Impact analysis of Internship towards attainment
4. Comparative analysis of CO attainment for Design / Manufacturing / Thermal domains
5. Proposed methodologies to set target attainment for CO i) Design Domain and ii) Manufacturing Domain
6. Student's performance in Co-Curricular and Extra Curricular Activities
7. Faculty Progress
8. Budget utilization (AY2021-2022)
9. Continuous Improvement – department progress

1. REVIEW OF PREVIOUS MEETING (4th PA&QIC)

Dr.P.Sureshkumar, ASP/Mech. presented the progress and action taken for the suggestions given in 3rd PA&QIC meeting held on 11.02.2022. The action taken was briefed by him and the same was presented in detail by the corresponding process owners.

Item No.	Comments given in 4 th PAQIC	Actions Taken
Impact Analysis of Internship and Industry Training	Encourage students to participate in internships and in-plant training at all of the organizations with which we have signed Memorandums of Understanding.	7 out of 27 organizations where students received internship and in plant training
	Improve the stipend based internship opportunities to the students	3 Students completed internship with stipend (Zoho, Intellipaat, Bengaluru etc.,) (Proof is attached as an Annexure I)
	Incorporate quantitative measures in outcome attainment.	Internship outcome attainments were calculated as CO1: 3 & CO2:3 for the Batch 2018-2022
Analysis and Action Taken for Institute and Industry Interaction	MoU's supporting entrepreneurship activities can be signed in future for the benefit of students	MoU signed in Rajapalayam Raju's college for emphasizing on Entrepreneurial activities through IIC. https://www.ritrjpm.ac.in/images/pdf/2021-2022/Rajus_CLG_MOU.pdf
	Few of the MoU's which are in expiry should be renewed	Steps taken and discussed in 11 th CAC meeting held on 01.07.2022 (Friday). The website content will be revised before the next academic year through IIC.
	Domain specific industries MoU's can be signed in future for the efficient outcomes	Plan for signing domain specific companies and R&D Lab, at present Design: 07, Manufacturing: 09, Thermal: 02
Students' Participation in Inter-Institute Events	Formulate the team members consist of senior and junior students in various competitions in order to disseminate knowledge and exposure among students.	Student group formed in SAE driverless vehicle Competition. M.Kirubakaran (III Year) and R.Thiyanesh Sakthi (II Year) & P.J. Jaswin Shree Raam (II Year)
Training and Placement – Quantitative and	Calculate the Median and Average salary of each batch student salary as it gives various insights into quality of	Calculated for (Batch - 2018 -2022) Average Salary 2.43 LPA Median Salary 2.25 LPA Median Salary (Batch - 2017 -2021)

Qualitative Analysis	the placement.	2 LPA
	Maintain knowledge repository of questions asked in interviews.	Interview questions were collected from the students and maintained in the department.
Analysis of Faculty Contributions to the Improvement of Students' Learning	Map the faculty participations to the subject and to extract more outcomes.	Online course and ATAL FDP leaned content are used in their respective subjects.
	Prepare monogram and chart and circulate the same to the students.	The monogram and chart preparation was discussed in DRM and it will be done this semester
	Get critiques and reviews for the innovative practices from the various stake holders.	Discussed in Department Review Meeting / 2021-22 / Even Semester / 02 and Domain wise course were selected.
	Appoint adjunct faculty for the department	Appointed Prof. L.Vijayaraghavan, Retired Professor, IIT Madras, Chennai as an adjunct faculty and he have taken 10 hours through online and offline mode.
Review of Faculty R&D Progress	Appreciated the faculty R&D progress and outcomes generated.	Best practices will be continued.
	Avoid publishing the papers other than Scopus and SCI (E) indexed journals and enhance the number of quality publication indexed in SCI (E).	Faculties are publishing only Scopus Indexed (2 Papers) and SCI (E) Indexed Journals (7 Papers) during this period. (Proof is attached as an Annexure II)
	Involve the student members to take part in research proposal submission.	Student members were included in IE (I) proposal and MSME proposal (Mr.M.Bharath, III Mechanical 'A' 953619114011).
Continuous Improvement – Department Progress	Formulate peer review committee to ensure the quality of e-learning resources and e-lectures to be uploaded in e-sharing forums.	Nominated the committee which consists of senior faculty members and Criterion 2 faculty in charges.
	Congratulated the department on achieving a higher placement index and suggested to improve average salary.	It has been proposed to increase the average salary through improved placement record.

Appreciated department faculty members for their contributions towards process refinement and its implementation.	Best practices will be continued.
Concentrate more on student progression towards higher education and entrepreneurial culture.	GATE coaching classes were conducted for the students on 30.03.2022 (Two classes), 31.03.2022 (Two classes), 28.04.2022, 06.05.2022, 11.05.2022, and 12.05.2022 & entrepreneurial awareness program conducted through IIC (https://www.ritrjpm.ac.in/images/mechanical/2021-2022/2021-22_IIC_Consolidated_Report.pdf).

Suggestions & comments by PAQIC member:

- It is suggested to specify that the action taken must be quantifiable and qualitative.

2. TEACHING AND LEARNING:

Presenter: Mr.G.Prabu ram, AP (SG)/Mechanical

Briefed the following new initiatives taken for quality improvement in teaching and learning:

1. Revamping of Question Paper Quality Assessment (Facilitation Index)
2. Revamping of Evaluation Quality of Assessment (Facilitation Index)
3. Rubrics are used to evaluate projects (AICTE Examination Reforms Policy, Page 49-51).

Based on Individual Assessment and Group Assessment.

Shared the inferences received from the feedback analysis [Students/Recruiters/Parents/Alumni] and listed out the action taken as follows:-

1. It was noted that all of the process owners for teaching and learning received more than 82% positive feedback.

Feedback about the outcome of teaching learning was received from the employee and is as follows:

1. It needs to improve its ability to use modern tools.
2. Developing or creating a solution for a complex problem may be inculcated.

The action taken for employer feedback is as follows:

1. Various new software and modern tools are used by faculties and students in their work.
2. Students are motivated to do industrial projects to provide solutions to real time problems.

Inferences from the recruiters' feedback about the outcomes of teaching and learning are as follows:

1. Needs to improve communication and soft skills.
2. Technical competence can be improved.

The action taken for recruiters' feedback is as follows:

1. GATE Classes are arranged for bright students to increase their technical competence.
2. With the help of the English department, communication skills and soft skills are being improved.

Parents' Feedback

1. Parents were requested to conduct the special classes for slow learners during the evening time.
2. Faculty Hostel visit help the students for clearing there subjects in the semester exams.

The action taken for parents' feedback is as follows:

1. Special classes were conducted for slow learners during the evening time.
2. Study hour monitoring duty was implemented in the hostel.

Alumni Feedback analysis is tabulated below.

Technical	A Sec (out of 5)	A Sec in %	B Sec (out of 5)	B Sec in %	AVG (2018-22)
Using information technologies and search strategies to access information	4.36	87.2	4.4	88	87.6
Critically analysing and evaluating ideas/ information	4.45	89	4.3	86	87.5
Examining problems and integrating/ synthesizing ideas/information from multiple sources	4.28	85.6	4.3	86	85.8
Constructing clear, precise and accurate arguments/ statements/ reports	4.36	87.2	4.38	87.6	87.4
Generating innovative solutions	4.47	89.4	4.26	85.2	87.3
Problem solving skills	4.43	88.6	4.28	85.6	87.1

Suggestions & comments by PAQIC members:

- It is suggested to check the Facilitation Index (FI) value specified in the Assessment of Question Paper Quality template and get the feedback from the exam cell in this regard.
- The Discrimination Index (DI) of the question paper should be removed from the Assessment on Quality of Evaluation.
- Advised to include more responses received from parents' feedback and analysis (get the inferences and take the necessary action) of the same.

- Specified to include the feedback received from the class committee and course committee and any necessary action taken.
- It is proposed to consolidate the list of curriculum gaps identified in the previous semester and ensure that the gap is addressed through appropriate instructional methods either in the same semester or the following semester.
- It is recommended to give a specific certificate for those students who have completed value-added courses.
- It is suggested to assess the value added course and include in the PO and PSO attainment calculation.

3. **IMPACT ANALYSIS OF INTERNSHIP TOWARDS ATTAINMENT**

Presenter: Mr.S.Kathiravan, AP/Mechanical

- Presented the academic wise internship details (2018-19 : 39, 2019-20: 49, 2020-21: 35, 2021-2022:102)
- Listed out the Internship details i) Placement with stipend : Inellipat:-Two students/ Rs.22,000 PM. & Zoho: One Student/Rs. 15,000 PM
ii) MoU companies Internship:- The Ramco Cements (12 students), Medsby Health care solutions (19 Students) and Eleation (3 Students)
iii) Project:- Gowri House Metals – 6 students & Rajapalayam Mills – 4 students
- Shown the CO attainment calculation for the Batch 2018-2022 (CO1-3 and CO2-3)
- Classified the internship into Manufacturing /Campus Ambassador / Artificial Intelligence and IoT / Industrial Engineering & Design Engineering and mapped with POs and PSOs.
- Presented the sample feedback received from the students about their internship.
- Listed out the activities carried out as follows:
 - Revision of ISO Form,06 – In-plant Training Progress Report has been in progress.
 - Internship mark statement, consolidated details and certificates has been sent to Anna University CoE office.
 - Letter has been sent to include Internship in listed course in Employability enhancement courses -Anna University Centre for Academics.

Suggestions & comments by PAQIC members:

- It has been proposed to include CO attainment in the calculations for POs and PSOs. Identify any difficulties that will occur during the calculation part of the forthcoming semester.
- It is suggested to ensure that the students are doing internships in the thrust area and enhancing their knowledge level.
- It is suggested to avoid the usage of the "In-plant Training" word in the current practice.
- It is proposed to observe and take the necessary action on the feedback collected from the students regarding internship..

- Sort internships by i) industry, ii) start-ups, and iii) software.

4. COMPARATIVE ANALYSIS OF CO ATTAINMENT FOR DESIGN / MANUFACTURING / THERMAL DOMAINS

Presenter: Dr.J.Jabinth, AP/Mechanical

- a. Assessment methods practiced in the last academic year.

Direct	Indirect
IAT I, II, III Assignments, Tutorials, etc., University Assessment (40% + 60%) Course End Survey (10% Weightage)	Exit Survey (2017-21 batch) Program Exit Survey (97) Alumni Exit Survey (15) Exit Survey (2018-22 batch) Program Exit Survey (75) Alumni Exit Survey (94)

- b. Total Number of subjects for which course Attainment calculated in Academic Year 2020 – 21 is 79 & 2021-22 is 80 and CES is collected by the respective subject faculty.
- c. Briefed the existing course attainment procedure & GPA calculation for target setting
- d. With the aid of bar chart compared the semester wise CO attainment for various batches and listed out the CO attained and not attained courses.
- e. Classified the courses into Design / Thermal / Manufacturing and General and listed out the number of attained and not attained courses.

DESIGN DOMAIN (Total – 19+2 Project)

Attained	17+2	12+1	6	1
Not Attained	2	2	0	1

THERMAL DOMAIN (Total subjects – 11)

Attained	8	5	3	0
Not Attained	3	2	0	0

MANUFACTURING DOMAIN (Total Subjects – 13)

Attained	13	9	4	1
Not Attained	0	0	1	0

GENERAL PAPERS (Total – 34)

Academic Year	2017-21	2018-22	2019-23	2020-24
Attained	32	22	18	12
Not Attained	2	6	4	4

- f. CO-PO Attainment for 2017 – 21 batch

- Total Number of subjects in Mechanical Engineering for Academic year 2017 – 21 is 79.
- Mapped subjects list
 - PO1 – 68, PO2 – 69, PO3 – 63, PO4 – 66, PO5 – 49, PO6 – 33, PO7 – 27
 - PO8 – 44, PO9 – 36, PO10 – 65, PO11 – 20, PO12 – 69
 - PSO1 – 40, PSO2 – 54, PSO3 – 40, PSO4 – 29

g. PO/PSO Attainment

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
2.19	1.84	1.68	1.67	1.77	1.72	1.64	1.49	1.88	1.63	1.91	1.48

PSO1	PSO2	PSO3	PSO4
1.49	1.45	1.37	1.56

h. Program Exit Survey 2017-21 batch (97) & AES (15)

Program exit survey is also collected via online mode, which contains 25 questions and are consolidated, mapped to get attainment for each POs.

AES – 2.53

PES

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
2.64	2.61	2.63	2.626	2.638	2.597	2.601	2.599	2.604	2.61	2.642	2.625

PSO1	PSO2	PSO3	PSO4
2.55	2.69	2.61	2.69

Suggestions & comments by PAQIC members:

- ✓ Suggested to check the attainment value of the course C106 Engineering Graphics and double check all other courses also.

5. PROPOSED METHODOLOGIES TO SET TARGET ATTAINMENT FOR CO I) DESIGN DOMAIN AND II) MANUFACTURING DOMAIN

Presenter: (i) Mr.J.Jerold John Britto, AP (SG)/Mechanical
(ii) Mr.S.Maharajan, AP/Mechanical

- Explained the existing methodology for course attainment
- For Manufacturing Domain proposed a new approach for target setting mechanism

S.No.	Parameters	Allocated Weightage (%)			
		Method 1	Method 2	Method 3	Method 4
1.	Number of Times Subject handled	10%	10%	10%	10%
2.	Faculty Experience	30%	20%	10%	10%
3.	Course Quality	20%	30%	20%	10%
4.	Student Quality	40%	40%	60%	70%

S.No.	Parameters	If Method 2 Selected	Proposed Criteria
1.	Number of Times Subject handled	10%	Less than 2 times handled – 5% weightage (1.5) Equal to 2 times – 7.5% weightage (2.25) Equal to 3 and above 3 times handled – 10% weightage (3)
2.	Faculty Experience	20%	Less than 3 years' experience - 8% weightage (1.2) Equal to and greater than 3 and less than 5 Years - 12% weightage (1.8) Equal to and greater than 5 Years - 20% weightage (3)
3.	Course Quality	30%	Toughness level High – 15% Weightage (1.5) Toughness level Medium – 20% weightage (2) Toughness level Low – 30% weightage (3)
4.	Student Quality	40%	Overall Grade point average (GPA) of previous semesters results (In case III semester subjects, consider student's performance in I & II Semester and HSC marks)

- Presented the sample calculation for ME8351 Manufacturing Technology I (2022-23 (Odd)) Subject Faculty: Mr.S.Maharajan & Mr.M.Lakshmanan CO Target: 2.48
- Presented the rubrics for Subject Quality. The value for subject quality is determined by the difficulty level of the subject based on theoretical/problem-solving/programming/lab exercises.

S.No.	Quality	Difficulty level	Weightage (in 3 point scale)
1.	No of CO – PO mapping 3	Very Low	3
2.	No of CO – PO mapping 5	Low	2.5
3.	No of CO – PO mapping 7	Medium	2
4.	No of CO – PO mapping 10	High	1.5

S.No.	Quality	Difficulty level	Weightage (in 3 point scale)
5.	No of CO – PO mapping 12	Very High	1

- Design domain coordinator explained a new approach for target setting mechanism.
- Expected Proficiency (EP) and Expected Level of Attainment (ELA) (%)
- Shared the sample calculation for Course Name: ME8594 - Dynamics of Machines

Batch of the Student	O	A+	A	B+	B	U	Student Count
2017 - 21	0	3	19	18	56	20	116
2018 - 22	0	38	28	21	11	0	98
2019 - 23	3	38	5	1	0	0	47
Sum of Students	3	79	52	40	67	20	261
Cumulative Percentage (%)	1.15	31.42	51.34	66.67	92.34	100	-
Attainment Level (%)	1.44 (1.5)	39.27 (40)	64.18 (65)	83.33 (85)	115.42 (100)	125 (100)	-
Expected Proficiency (EP)			A				-
Expected Level of Attainment (ELA) (%)			65				-

Suggestions & comments by PAQIC members:

- ✓ Suggestion for Manufacturing Domain: Number of Times Subject handled – 10 %
Faculty Experience – 10 %, Course Quality – 40 % and Student Quality – 40 %
- ✓ The design domain coordinator's suggested methodology should be verified and implemented in one course.
- ✓ It is proposed to consider students' +2 PCM average for first year course outcome attainment calculation and previous semester course performance for higher semester CO attainment calculation.

6. STUDENT'S PERFORMANCE IN CO-CURRICULAR AND EXTRA CURRICULAR ACTIVITIES

Presenter: Mr.S.Valai Ganesh, AP/Mechanical

- Presented the student participation details co-curricular activities during January to June 2022
 - Training/Competitions – 26 , Workshop – 15, Online Course – 22 , Quiz,IV/IPT/Others – 35 and Paper Presentation/Conference - 19
- Highlighted the Outside Tamil Nadu Participations

- Two students presented Papers in International Conference on Sustainable Materials, Manufacturing & Industrial Engineering (ICSMMIE-2022) organized by Siddaganga Institute of Technology, Karnataka during July, 2022,
- II Mech Student attended an event on "Analytico, Who's The Boss and Brainstorm" organised by VNIT, Nagpur during 24-27 Mar., 2022
- 14 IV Mech Student completed their "Advanced Diploma in Safety course practical" conducted by the TÜV Rheinland NIFE Academy during 24-31 Mar., 2022 in Warangal, Telangana.
- 7 II Mech, 12 III Mech and 1 IV Mech students participated in a "NVIDIA Global Technology Conference" organised by NVIDIA, USA during 21-24 Mar., 2022
- 1 IV Mech student participated in the Online International Seminar on "Changing Paradigms of Global Supply Chains" conducted by EM Normandie Business School, France on 11 Mar., 2022
- Listed out the student achievements
 - The Hon'ble Chief Minister Shri M. K. Stalin congratulated our students R. Dinesh and Soumya Souvik Khuntia of III Mech received Bronze Medal in "Robot System Integration Skill" during National Finals and received their medals at New Delhi during January 2022 on 14 Mar., 2022
 - S. Jehaveerapandian, G. Dinesh Kumar, III Mech received I Prize in "Circuit Design and Debugging" organized by the Department of Mechatronics, Thiagarajar College of Engineering, Madurai on 28 May, 2022
 - R. Dinesh, R. Shyam Sundar, III Mech received II Prize in "PLC Circuit Design Challenge" organized by the Department of Mechatronics, Thiagarajar College of Engineering, Madurai on 28 May, 2022.
- Student Publications and Grant Details
 - One Copyright work registered in Artificial Intelligence in Automotive Systems
 - Two Patents were filed during this period
 - Received a grant of Rs. 1,82,800/- from MSME and Rs. 31,970/- from IEI
 - 19 Papers were presented in National and International Conference conducted across India
 - Extra-Curricular Activities (Annual Sports Day 2021-2022)

S.NO.	Game	Winner - Gold	Runner - Silver	Runner - Bronze
1.	KABBADI	02 IV Mech	01 III Mech	-
2.	FOOTBALL	01 II, III and IV Mech	02 III, IV, and 01 II Mech	-
3.	CRICKET	04 II and III Mech	02 I and 04 IV Mech	-
4.	VOLLEYBALL	01 IV Mech	01 IV Mech	-
5.	HOCKEY	01 I, II and IV Mech	02 II, 01 III and IV Mech	-

S.NO.	Game	Winner - Gold	Runner - Silver	Runner - Bronze
6.	BALL BADMINTON	03 I Mech	01 II and 03 III Mech	-
7.	BADMINTON (DOUBLES)	-	01 III Mech	-
8.	TABLE TENNIS	-	01 III Mech	-
9.	JAVELIN	II MECH	-	II MECH
10.	DISCUSS	II MECH	-	I MECH
11.	HAMMER THROW	II MECH	IV MECH	
12.	4 X 100 MTS RELAY	III MECH	III MECH	I MECH
13.	4 X 400 MTS RELAY	III MECH	III MECH	III MECH
14.	200 MTS	-	III MECH	-
15.	5000 MTS	-	III MECH	I MECH
16.	LONG JUMP	-	III MECH	-
17.	SHOT-PUT	-	IV MECH	-
18.	TRIPLE JUMP	-	III MECH	I MECH
19.	100 MTS	-	-	III MECH

Suggestions & comments by PAQIC members:

- Suggested to assess the student's participation effectiveness in co-curricular activities through a quiz and consider it for PO attainment.
- At around 20% of students' participation may be considered for attainment calculation.
- Proposed to study the challenges that will occur in the attainment calculation and project them in the next meeting.

7. FACULTY PROGRESS

Presenter: Dr.S.Godwin Barnabas, ASP/ Mechanical

- Presented the faculty participation in Webinar (15) / STTP (1) / FDP (11), acted as a resource person: within RIT – 3 Other Institutions – 14, attended in Human Value Course: UHV refresher I (8), UHV in Technical Education (16)
- Listed out the industrial visit carried by the faculty members
 - Dr.S.Godwin Barnabas ASCP/Mech carried out the industry visit in “G.P.Leather” (unit 1 and 2) on 10th and 13th January 2022, Poraiyur, Puducherry.
 - Dr.S.Godwin Barnabas ASCP/Mech carried out the industry visit in “Sun beam generators” on 11th and 12th January 2022, Pathhukannu, Puducherry.
 - Dr. P. Sureshkumar, ASP/Mech, Mr. M. Ashok Kumar, AP(SG)/Mech visited “HLL Life Care Limited, Akkulam Factory, Thiruvananthapuram, Kerala” as a part of Faculty Industrial exposure programme during 14-16 Feb., 2022.

- Mr. M. Sivagaminathan alias Balaji, AP(SG)/Mech, Mr. C. Gururaj, AP(SG)/Mech, Visited **"KeraFed-Kerala Kerakarshaka Sahakarana Federation, Ltd.(A Government of Kerala Enterprise)"** as a part of faculty - industry exposure programme during 14-16 Feb., 2022..
- Dr. S. Godwin Barnabas, ASP/Mech, Dr. J. Jabinth, AP/Mech, Mr. G. Prabu ram, AP(SG)/Mech, Mr. P. Pavithran, AP/Mech, along with lab technicians visited **"Rajapalayam Mills Ltd. and Rajapalayam Spintext training session on Karakuri Kaizen"** on 04 Mar., 2022.
- Mr. N.L. Sujin, AP / Mech. Visited **"V V Goldtech Cold Storage, Madurai"** as a part of Faculty Industrial exposure programme during 25-27 Jan., 2022.
- Presented the publication details: Journal (9) / National Conference (1) / International Conference (12)
- Updated the progress of the funded project from DST SSTP, RuTAG IIT Madras, and IE(I) India.
- 1 Patent filed, 1 copyright registered, 2 FER received for the already filed patents and 3 projects were selected for funding under MSME Innovative Scheme.
- Presented the Awards / Achievements / Recognition achieved during this period
- Dr. S. Rajakarunakaran, Prof. & Head /Mech. Acted as an expert member for **"Mock NBA audit"** conducted at NSR Institute of Technology, Sontyam, Visakhapatnam on 26 Jan., 2022, SSM Institute of Engineering and Technology, Dindigul during Feb., 12 – 13, Anna University chennai during Mar., 22 – 23, 2022.
- Dr. P. Sureshkumar, ASCP /Mech. Acted as a expert guest for evaluating **"Self Assessment Report"** conducted at Anjalai Ammal Mahalingam Engineering College, Thiruvavur on 21 March, 2022.
- Mr. C. Gururaj, AP (SG)/Mech, nominated as an evaluator for evaluating the ideas that have been received for **"Smart India Hackathon - 2022 Junior"** initiated by MoE's Innovation Cell, The Ministry of Education, Government of India on 21 May, 2022.
- Mr. M. Sivagaminathan alias Balaji, AP (SG) / Mech. Received **"Best Library User"** award from Ramco Institute of Technology 04 June, 2022.
- Mr. N.L. Sujin AP/Mech received the **"Chapter Stalwart Award"** during the 12th chapter installation function of the ISHRAE Madurai chapter on 27 Mar., 2022.
- Mr. R. Arun Kumar, AP/Mech is certified as an **"NPTEL Motivated Learner"** for the period Jan.,-Apr., 2022.

Suggestions & comments by PAQIC members:

- ✓ PAQIC members suggested to prepare faculty wise list of participation in ATAL FDP / STTP / Webinar / etc.,

- ✓ Proposed to update the NBA P15 file regarding faculty interaction outside the world.
- ✓ Advised to concentrate on the consultancy work.

8. **BUDGET UTILIZATION (AY2021-2022)**

Presenter: Mr.M.Sivagaminathan alias Balaji, AP(SG)/Mechanical

- The department NBA criteria 10 coordinator was entrusted with the responsibility to present the budget utilization for this academic year 2021 – 2022. The following table has the details of department budget allotted and expenses for the academic year 2021 – 2022.

Budget Allotted	Rs. 22,36,493.00
Budget expenses	Rs. 22,21,026.63
Budget utilization %	99.3%

- The presenter detailed the various heads under which the expenses are classified and explained the steps taken to ensure the maximum utilization of the budget.

Suggestions & comments by PAQIC members:

- The PAQIC members called students as primary stakeholder and suggested to seek budget requirement from students for developing facilities which will be useful to them while doing projects.
- It is proposed to show the fund requested, fund allotted, and fund utilised in the presentation.

9. **CONTINUOUS IMPROVEMENT – DEPARTMENT PROGRESS**

Presenter: Mr.L.Karthikeyan, AP/Mech.

- Presented the academic audit process and actions taken thereof during the period of assessment
- Highlighted the process refinement in Academic Audit.
- Presented ISO form & process refinement leads to academic improvements
 - Answer script Evaluation form
 - Faculty identification for critiques on innovative practices and template for feedback comments
 - Course curriculum meeting on 18.04.2022– “Advances in Manufacturing”
 - LaTeX for Project report
 - NBA cycle –II Plan of activities
- Projected the improvement in Placement, Higher Studies and Entrepreneurship

Item	(2016 -17)	(2017-18)	(2018-19)	(2019-20)	(2020-21)	(2021-22)
Total No. of Final Year Students (N)	71	141	143	123 (111 willing)	118 (103 Willing)	98 (90 Willing)

Item	(2016 -17)	(2017-18)	(2018-19)	(2019-20)	(2020-21)	(2021-22)
No. of students placed in companies (x)	45	98	91	82	99	90
No. of students admitted to higher studies with (y)	00	07	06	04	3	2
No. of students turned entrepreneur (z)	01	02	06	06	Nil	Nil
$x + y + z$	46	107	103	92	102	92
Placement Index : $(x + y + z)/N$	64.78	75.88	72.03	74.79	86.44	93.87
Average placement: $(P1 + P2 + P3)/3$	70.89			85.03		

- Presented the Placement Statistics of the Academic year 2021 – 2022 – Batch 2018-2022

S.NO.	Criteria	Quantitative Detail
1.	Total Number of students registered for Placement	90
2.	Total Number of students Placed	90
3.	Total Number of Offers	93
4.	Number of On campus Recruitment drives conducted	23
5.	Average Salary	2.43 LPA
6.	Median Salary	2.25 LPA
7.	Median Salary (Batch - 2017 -2021)	2 LPA

- Showed the improvements in students admission details

Item	Particular	2020-21 (120)	2021-22 (60)
Regular students	No. of students admitted	40	57 (95%)
	Opening score	162.5	184.21
	Closing score	80	132.13
Lateral Entry students	No. of students admitted	07	45
	Opening score	88.71	94.52
	Closing score	62.2	59.96

- Listed out the Alumni Interaction series – Completed programs

<i>Nature of session</i>	<i>Alumni – Resource person</i>
Project review member (Final Project)	Mr.S.Nithish, Associate design, Radiant design & Ms.P.Keerthana, Associate consultant, Krysalis Consultancy services
Project review member (Mini Project – Design & fabrication)	Mr.R.Vishnusankar, Assistant Manager (Mechanical), The Ramco cements Ltd.,

Suggestions & comments by PAQIC members:

- Suggested to refer to the NPTEL forum and validate the allocation of domain coordinators.
- Receive input from the distinguished alumni during the formation of new value-added courses. The proposed value-added course content should be in the AICTE prescribed minor courses.
- Use concept mapping to improve teaching and learning.
- Ensure that the dissemination of Vision/Mission/POs/PSOs/PEOs takes place in the extended laboratories.
- Motivated faculty members to become acquainted with their thrust areas and directed faculty members to guide project work in their specialised thrust areas.
- Appreciated the faculty members, Mr.R.Arun Kumar, AP/Mech. and Mr.G.Prabu ram, AP (SG)/Mech., who have contributed to the coordination of MSME project proposal submission.
- Encouraged the faculty members to submit the project proposals along with the student community.
- Periodically review the progress of UHV/NITTR course faculty members.
- Appreciated the regular conduct of the Extra Mural Lecture series (National level – Dr.J.Jabinth AP/Mech.) and suggested including a resource person from the industry, which may support getting a better CII ranking.
- Insisted on periodically updating the Online Course/Industrial Visit/Internship/Student Participation and Outcome of the Student Project.
- Applauded the progress in social media promotion (led by Mr.S.Valai Ganesh, AP/Mech. and Mr.R.Prabhakaran, AP/Mech.) and advised the department to disseminate special occurrences through appropriate press releases.
- Mr.M.Sivagaminathan alias Balaji, AP (SG)/Mech. was appreciated and recommended to meet with other department criteria 10 coordinators to discuss about the budget utilization.

V. Suresh
Prepared by 16/8/22

NBA Coordinator/Mech.

Reviewed by
(HOD/Mech)

Approved by
Principal



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering

Internship with stipend received during AY 2021-2022

Annexure I



Software Solutions Pvt. Ltd.

AMR Tech Park 3 Ground Floor,
Tower B, Hongsandra Village,
Bommanahalli, Hosur Road,
Bengaluru-560068
(M): +917022374614
www.intellipaate.com

Date: October 13, 2021

To,

VASU KRISHNA,
Subject: Internship Letter

Dear Vasu Krishna,

In reference to your application we would like to congratulate you on being selected for internship with IntelliPaat Software Solutions Pvt. Ltd. Your work is scheduled to start from 18th October, 2021 for a period of 1 year. During this period, you will get paid Rs. 22,000/month (Rupees Twenty Thousand Only) and you will be working as an 'Business Development Trainee'. The technical platform and job role will be shared with you by your manager on the day of your joining.

Note:

- Number of working days is 6 days in a week.
- Your first month salary will be deposited along with your 4th month salary.
- During this internship you are eligible to get 200,000/- Rupees per annum as incentive based on your performance.
- During your internship, you are expected to be disciplined and sincere towards your job responsibility.
- Based on your performance, PPO will be released before the completion of your internship at IntelliPaat.
- After the successful completion of 1 year internship period, you are eligible to get 900,000/- Rupees per annum as a gross pay.
- You need to serve 1 month of notice period without fail, or else the management of IntelliPaat can hold your salary/internship letter / relieving letter after joining.
- Minimum working duration must be 9 hours including one-hour lunch/dinner break.

Again, congratulations and we look forward to working with you.

Yours truly,
For IntelliPaat Software Solutions Private Limited



Dev Bisht
Director - Human Resources

IntelliPaat Software Solutions Pvt. Ltd.
AMR Tech Park 3, Ground Floor, Tower B, Bommanahalli, Hosur Road, Bengaluru, Karnataka-560068



Zoho Corporation Private Limited

Plot 140, 151, Estancia IT Park, Vallancheri,
Chengalpattu District, Tamilnadu, 603 202.

Ph: +91-44-6744 7070

www.zohocorp.com

SEZ Unit

Date: 29-Nov-2021

To

Mr.RAGHURAM K.

46AS/12,

KATHIRESAN KOVIL STREET,

KOVILPATTI-628502, TAMIL NADU.

Dear Mr.RAGHURAM K,

OFFER OF EMPLOYMENT

We are pleased to offer you employment for the position **TECHNICAL SUPPORT ENGINEER** with **ZOHO CORPORATION PRIVATE LIMITED**.

INTERNSHIP AND STIPEND

You are expected to do the final semester project of your curriculum in our organization. We expect you to work on the project on a full time basis for a period of 5-6 months. During this period you will be paid a monthly stipend of **Rs.15000/- (RUPEES FIFTEEN THOUSAND ONLY)**. The following offer is valid, subject to successful completion of your project.

(Note: The above may not apply to you if your college does not permit internships)

REMUNERATION

Your annual Cost to Company will be **Rs.480000/- (RUPEES FOUR LAKH EIGHTY THOUSAND ONLY)**. The break-up of your gross salary and information specific to bonus and gratuity are set out in Annexure A. Salary will be paid by the last day of each month. All additional benefits that Zoho currently provides employees are set out in Annexure B.

DATE OF JOINING

Your date of appointment is effective from your date of joining after successful completion of your curriculum.

PROBATION

You will be on probation, at a minimum, until completion of the performance review cycle that immediately follows completion of six months from your date of joining, provided that your performance is determined to be satisfactory. If your performance is not satisfactory, your probation may be extended until your performance is determined to be satisfactory. Upon completion of the probation period you will be confirmed on the rolls of the company.

SALARY REVISION

Revision to your compensation will be after one year from the date of joining, subject to satisfactory completion of the probation by you. Zoho operates a Pay for performance Policy and any salary revision will take your performance into account.

Corporate Identification No: U40100TN2010PTC075961

e-mail ID: hr-team@zohocorp.com



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

Accredited by NAAC & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering

JOURNAL PUBLICATIONS AY 2021-2022

Annexure II

S.No.	Title of the paper	List of authors	Journal Name
1	Sustainable Development of Carbon Nanocomposites: Synthesis and Classification for Environmental Remediation	V.Dhinakaran, M.Varsha Shree, P.Suresh Kumar, T.Jagadeesha, L.Natrayan, M.Ravichandran, Prabhu Paramasivam	Journal of Nanomaterials
2	Additive Manufacturing of Biopolymers for Tissue Engineering and Regenerative Medicine: An overview, Potential applications, Advancements and Trends	V.Dhinakaran, M.Swapna Sai, P.Suresh ¹ Kumar, T.Jagadeesha, M.Ravichandran, Wubishet Degife Mammo	International Journal of Polymer Science
3	Energy and exergy analysis of conventional acrylic solar still with and without copper fins	Sivakumar Vaithilingam, Vimala Muthu, Muthu Manokar Athikesavan, Asif Afzal & Ravishankar Sathyamurthy	Environmental Science and Pollution Research
4	Enhancement of mechanical properties on novel friction stir welded Al-Mg-Zn alloy joints reinforced with nano SiC particles	L.Natrayan, M. Ravichandran, V.Dhinakaran, P.Sureshkumar, T.Jagadeesha, R. Suryanarayan, Wubishet Degife Mammo	Journal of Nanomaterials
5	Crystallographic orientation and wear characteristics of TiN, SiC, Nb embedded Al7075 composite	V. Chakkravarthy, M. Lakshmanan, P. Manojkumar, R. Prabhakaran	Materials Letters

S.No.	Title of the paper	List of authors	Journal Name
6	Influence of nano-graphite on dry sliding wear behaviour of novel encapsulated squeeze cast Al-Cu-Mg metal matrix composite using artificial neural network	Natrayan L, Ravichandran M, Dhinakaran Veeman, Sureshkumar P , Jagadeesha T, Wubishet Degife Mammo	Journal of Nanomaterials
7	Sustainability and Environmental Impact of Ethanol, Oxyhydrogen addition on Nano-coated Gasoline Engine	Sambandam Padmanabhan, K Giridharan, Balasubramaniam Stalin, V Elango J.Vairamuthu, P Sureshkumar , Leta Tesfaye Jule, Ramaswamy Krishnaraj	Bioinorganic Chemistry and Applications
8	Electrochemical corrosion and tribological behaviour of AA6063/Si3N4/Cu(NO3)2 composite processed using single-pass ECAPA route with 120° die angle	P.Sureshkumar , T.Jagadeesha, L.Natrayan, M.Ravichandran, Dhinakaran Veeman, S.M.Muthu	Journal of Materials Research and Technology
9	Investigation of Friction Stir-Welded B4C Particles-Reinforced Copper Joint: Mechanical, Fatigue, and Metallurgical Properties	M. Vetrivel Sezhian, G. Chakravarthi, K. Giridharan, Balasubramaniam Stalin, B. Yokesk Kumar, P. Sureshkumar , J. Vairamuthu, and Ramaswamy Krishnaraj	Advances in Materials Science and Engineering
10	Microstructural analysis and mechanical behaviour of copper CDA 101/AISI-SAE 1010 dissimilar metal welds processed by friction stir welding	K. Giridharan P. Sevvil B. Stalin M. Ravichandran P. Sureshkumar	Materials Research Ibero-american Journal of Materials
11	Water quality analysis of solar still distillate produced from various water sources of El Oued region Algeria	Mohammed El Hadi Attia, M. Mohamed Thalib, Srinivasan Kumar, Asif Afzal, Sivakumar Vaithilingam , Ravishankar Sathyamurthy, A. Muthu Manokar	Desalination and Water Treatment
12	Evaluation of solid particle erosion and electrochemical corrosion of plasma sprayed WC/8YSZ coating on SS316	S Maharajan , F Michael Thomas Rex, D Ravindran, S Rajakarunakaran	Surface Topography: Metrology and Properties
13	Prediction of Temperature Distribution in Three Dimensional Solid Objects using COMSOL and Python	S.Maharajan, G.Prabu ram , G.Sakthi Priya, K.Lingadurai	Materials Today: Proceedings

S.No.	Title of the paper	List of authors	Journal Name
14	Role of modern technologies and Internet of things in the field of solid waste management	S.Godwin Barnabas,S.Valaiganesh,S.Rajakarunakaran	International journal of computer communications and control
15	Analysis of Recyclable Waste and Performance Indicator Measures of Solid Waste Details In Madurai City	S.Godwin Barnabas,K.Arun Vasantha Geethan	Fresenius environmental bulletin
16	Vital Role of Recent Technologies in Sorting Different Solid Waste Materials in Recycling Process	S.Godwin Barnabas,K.Arun Vasantha Geethan	Fresenius environmental bulletin